



ALLEN
CAREER INSTITUTE
KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
14-03-2021

JEE(Main + Advanced) : NURTURE PHASE-I, I(B), NURTURE LIVE PHASE-I & I(B)

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

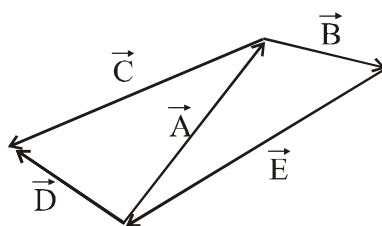
1. A wave is propagating along x-axis in a medium, so that displacement y of particle at any position x , at any time t is given by : $y = \frac{a}{t^2} e^{-\left[\frac{t}{b} - \frac{x}{c}\right]}$; where a , b and c are constants. Choose the

CORRECT option(s).

- (A) dimension of b is [T] (B) dimension of a is [LT²]
 (C) dimension of c is [L] (D) $e^{-\left(\frac{t}{b} - \frac{x}{c}\right)}$ is dimensionless
2. Which of the following three physical quantities can be taken as fundamental physical quantities.
 (A) force, mass and velocity (B) energy, force and area
 (C) energy, mass and velocity (D) length, acceleration and frequency

Space for Rough Work

3. A vector $\vec{A}$ of magnitude 3 units is added to another vector $\vec{B}$ which has magnitude 6 units but has random direction anywhere in the same plane. When $\vec{B}$ has such a direction that angle made by resultant with $\vec{B}$ is largest
- (A) magnitude of resultant of $\vec{A}$ and $\vec{B}$ is $3\sqrt{3}$ units.
 (B) angle between $\vec{A}$ and $\vec{B}$ is 120° .
 (C) angle between $\vec{B}$ and resultant is 30°
 (D) angle between $\vec{A}$ & resultant is 60° .
4. Figure shows vectors $\vec{A}, \vec{B}, \vec{C}, \vec{D}$ & $\vec{E}$. Which of the following is/are **CORRECT** ?



- (A) $\vec{E} + \vec{D} - \vec{C} = -\vec{B}$ (B) $\vec{E} = -(\vec{A} + \vec{B})$ (C) $\vec{E} + \vec{D} - \vec{C} = \vec{B}$ (D) $\vec{D} - \vec{C} = \vec{A}$

Space for Rough Work

5. Which of the following expressions has magnitude unity .
- (A) $\frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|}$ (B) $|\hat{a} + \hat{b}|$ when angle between $\vec{a}$ and $\vec{b}$ is 120°
- (C) $\frac{\hat{a} - \hat{b}}{|\hat{a} - \hat{b}|}$ (D) $|\hat{a} - \hat{b}|$ when angle between $\vec{a}$ and $\vec{b}$ is 60°
6. In a fight for a cheese piece, Tom has position vector $2\hat{i}$, Jerry has position vector $2\hat{j}$ at an instant. Position vector of cheese considering Tom as origin is $3\hat{i} + 3\hat{j}$. Now mark **CORRECT** options.
- (A) Cheese is closer to Jerry than Tom.
- (B) Position vector of cheese from origin is $(5\hat{i} + 3\hat{j})\text{m}$.
- (C) Position vector of Tom considering Jerry as origin is $(2\hat{i} + 2\hat{j})\text{m}$.
- (D) Position vector of cheese considering Jerry as origin is $(5\hat{i} + \hat{j})\text{m}$.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Forces acting on a particle have magnitudes of 14, 7, and 7 N acts in the direction of vectors $6\hat{i} + 2\hat{j} + 3\hat{k}$, $3\hat{i} - 2\hat{j} + 6\hat{k}$ and $2\hat{i} - 3\hat{j} - 6\hat{k}$ respectively. The forces remain constant while the particle is displaced from point A (2, -1, -3) to B (5, -1, 1). Find the total work done (in Joule) on the particle. The coordinates are specified in meters.
- Minimum speed of a plane on runway for take off was found to depend on area of blade of wings, density of material of plane and acceleration due to gravity. If take-off speed of a model toy plane is 10 m/s then what will be take-off speed (in m/s) of a real plane identical in all aspects but scaled up 10^3 times. If take-off speed of real plane is 10^n then fill the value of n.

Space for Rough Work

3. According to Maxwell-distribution law, the probability function representing the ratio of molecules at a particular velocity to the total number of molecules is given by

$$f(v) = k_1 \sqrt{\left(\frac{m}{2\pi kT}\right)^3} 4\pi v^2 e^{-\frac{mv^2}{2kT}}$$

Where m is the mass of the molecule,

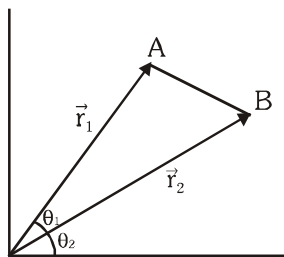
v is the velocity of the molecule,

T is the temperature

k and k_1 are constants.

If the dimensional formulae of k_1 is $L^a T^b \theta^c$ then find the value of $a + b + c$.

4. In a two dimensional motion of a particle, the particle moves from point A, position vector $\vec{r}_1$, to point B, position vector $\vec{r}_2$. If the magnitudes of these vectors are respectively, $r_1 = 3$ and $r_2 = 4$ and the angles they make with the x-axis are $\theta_1 = 75^\circ$ and $\theta_2 = 15^\circ$, respectively, then find the magnitude of the displacement vector.



Space for Rough Work

5. Two particles having position vectors (at $t = 0$) $\vec{r}_1 = (3\hat{i} + 5\hat{j})$ metres and $\vec{r}_2 = (-3\hat{i} - 1\hat{j})$ metres are moving with constant velocities $\vec{v}_1 = (4\hat{i} + 3\hat{j})$ m/s and $\vec{v}_2 = (7\hat{i} + \alpha\hat{j})$ m/s respectively. If they collide after 2 seconds, then the value of ' α ' is :
6. An insect starts from (2, 3, 4) and travels along $(\hat{i} + 2\hat{j} + 2\hat{k})$ with 15 m/s for 2 sec. Then it reverses its direction and travels with 3 m/s for 5 sec. If final position of the insect is (x, y, z), then find the value of (x + y + z).

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Resultant of two vectors $\vec{A}$ and $\vec{B}$ make an angle 53° with $\vec{A}$. Also the angle between $\vec{A}$ and $\vec{B}$ is 106° . If the magnitude of $\vec{A}$ is 5 N, what is magnitude of $\vec{A} + \vec{B}$ in N?
2. In a new system of units, unit of mass is 2 kg, unit of length is 4 m and unit of time is 2 second. How much joule is equal to 1 unit of energy in this new system?
3. The radius R, of the fireball produced by an atomic bomb explosion may be assumed to depend on the density ' ρ ' of the atmosphere, the time ' t ' elapsed since the explosion and the energy ' E ' released by the bomb. Deduce the formula for R by dimensional method. If your answer is $R = kE^{\frac{1}{x}}t^{\frac{2}{y}}\rho^{-\frac{1}{z}}$ where k is dimensionless constant. Fill the value of $\frac{x+y+z}{3}$.

Space for Rough Work

4. In a new system of units force (F), momentum (P) and mass (M) are taken as fundamental physical quantities. If the dimensional formula of power in new system is $F^a P^b M^c$ then find the value of $(a + b + c)$.
5. The angle of banking θ for a cyclist taking a curve is given by $\tan(\theta) = \frac{v^n}{rg}$, where v is velocity of cyclist, r is radius of curve and g is acceleration due to gravity. Then fill the value of n .
6. A particle moves in space such that its velocity and acceleration at certain point is given by $\vec{v} = (4\hat{i}) \text{ m/s}$ and $\vec{a} = (6\hat{i} + 2\hat{j} - 6\hat{k}) \text{ m/s}^2$. The magnitude of component of acceleration (in m/s^2) in the direction of velocity (known as tangential acceleration) is :

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
1. In an isotopic pair,
- (A) Number of protons are same
(B) Difference between neutrons and protons is same
(C) Number of neutrons are same
(D) Element is same
2. In which of the following sets of elements, the EA order is increasing but radius order is decreasing from 1st to 3rd element?
- (A) Mg, Al, Si (B) O, F, Ne (C) H⁻, H, He (D) Te, Se, S
3. Select the isoelectronic pair(s).
- (A) (CO₃²⁻, NO₃⁻) (B) (NH₄⁺, H₂O) (C) (ClO₄⁻, SiO₄⁴⁻) (D) (F₂, H₂S)
-

Space for Rough Work

4. Largest IE among the following is for.
(A) Fe (B) Fe^+ (C) Fe^{2+} (D) Fe^{3+}
5. Correct size order for d-block elements is/are :-
(A) $\text{Sc} > \text{Fe} < \text{Zn}$ (B) $\text{Cd} < \text{Y} > \text{Mo}$ (C) $\text{Cu} \approx \text{Ag} < \text{Au}$ (D) $\text{Sc} < \text{Y} \approx \text{La}$
6. For the process $\text{Mn}^{4+} \longrightarrow \text{Mn}^{7+}$, which of the magnetic moment value(s) is/are possible (including both Mn^{4+} and Mn^{7+} ions)?
(A) 5.90 BM (B) 3.87 BM (C) 2.83 BM (D) 1.73 BM
-

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Z_{eff} of F - Z_{eff} of Be = ?
2. $\frac{\text{Total number of f-block element}}{\text{Total number of s-block element}} = ?$
3. (Number of exchanges in d^8 configuration) $\times$ (number of unpaired electrons in Al)

Space for Rough Work

4. Maximum number of degenerate orbitals in $n = 5$ for multielectronic specie = ?

5. Minimum number of electrons in Cr with quantum numbers $m = 0$, $s = \frac{-1}{2}$.

6. $\frac{\text{Spin multiplicity for Mn}^{+4}}{\text{Number of exchanges in d-subshell in Mn}^{2+}} = ?$

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
 - The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
 - For each question, darken the bubble corresponding to the correct integer in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-

1. Numbers of elements from the following having higher IE than oxygen are.
He, Ne, N, Be, C, F
 2. Total number of d-block elements which are not transition elements:-
 3. Calculate total number of electrons having $\frac{n+l}{|m|} = 5$ for Zn atom?
-

Space for Rough Work

4. Find out total number of species which atomic radius less than K.
Li, Na, Ca, Sr, Ba, Cs, Cl, Rb, S
5. For La, period number – group number + number of unpaired electrons = ?
6. In how many number of pairs, first species is having higher EA than second?
[Na⁺, K⁺], [F⁺, Cl⁺], [P, As], [S, Se], [C, N], [Na, Mg]
-

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I : (Maximum Marks: 24)**

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. If $(x - k)$ is a factor of $x^3 + x^2 - k^2x + k - 2$, then
- (A) Sum of values of k is 1 (B) Product of values of k is -2
(C) Sum of values of k is -1 (D) Product of values of k is 2
2. If $4 - \sqrt{x} = \sqrt{2x - 68}$, then
- (A) Number of real solution is one (B) Sum of digits of x is 9
(C) x is a composite number (D) None of these
-

Space for Rough Work

3. If $\sqrt{\lambda} = \sqrt{4 + \sqrt{7}} - \sqrt{4 - \sqrt{7}}$, then
- (A) λ is a prime number (B) λ is an even number
 (C) λ is a composite number (D) λ is an odd number
4. If $a + b = 20$ & $a^3 + b^3 = 800$ & $ab = 2^x \cdot 3^y \cdot 5^z$, then
- (A) $x + y + z = 5$ (B) $x + y - z = 3$ (C) $x - y + z = 2$ (D) $x - y - z = 1$
5. If $a, b \in \mathbb{R}^+$, the minimum value of $\frac{8a}{b} + \frac{4b}{a}$ is $k\sqrt{2}$, then
- (A) k is divisible by 2 (B) k is divisible by 3
 (C) k is divisible by 4 (D) k is divisible by 5
6. If $a^3 + \frac{1}{a^3} = 18$, then the value of $a + \frac{1}{a}$ is not equal to
- (A) 1 (B) 2 (C) 3 (D) 4

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

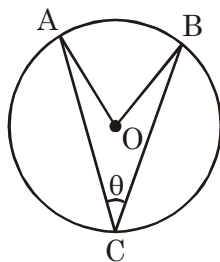
+										-									
●					●					●					●				
1	1	1	1	1	2	2	2	2	2	3	3	3	3	3	4	4	4	4	4
1	1	1	1	1	2	2	2	2	2	3	3	3	3	3	4	4	4	4	4
2	2	2	2	2	3	3	3	3	3	4	4	4	4	4	5	5	5	5	5
3	3	3	3	3	4	4	4	4	4	5	5	5	5	5	6	6	6	6	6
4	4	4	4	4	5	5	5	5	5	6	6	6	6	6	7	7	7	7	7
5	5	5	5	5	6	6	6	6	6	7	7	7	7	7	8	8	8	8	8
6	6	6	6	6	7	7	7	7	7	8	8	8	8	8	9	9	9	9	9
7	7	7	7	7	8	8	8	8	8	9	9	9	9	9					
8	8	8	8	8	9	9	9	9	9										
9	9	9	9	9															

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Product of real value(s) of x is k for the equation $(x + 1)(x + 2)(x + 3)(x + 4) = 120$, then the value of $\left| \frac{50}{k} \right|$ is
- The number of solution(s) of x satisfying the equation $\sqrt[3]{4} - 3 \cdot (2)^{\frac{1}{x+1}} + 8 = 0$ is

Space for Rough Work

3. In the given figure, if $\theta = 30^\circ$ and $\angle ABO$ is k° , then $\frac{30}{k}$ is



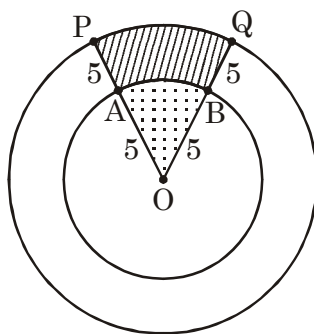
4. Polynomial $f(x)$ is divided by $(x - 2)$, then remainder become 3 and if $f(x)$ is divided by $(x + 2)$, then remainder become 5. If polynomial $f(x)$ is divided by $x^2 - 4$, then the remainder will be $px + q$. The value of $(q - p)$ is equal to
5. The sum of all possible real value(s) of x , satisfying the equation $(x^2 + 6)^{7-x} = 1$, is equal to k , then the value of $\frac{k}{2}$ is equal to
6. If $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then $n[(A - B) \cup (B - A)]$ is equal to

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If $x, y \in \mathbb{R}$ & $x \notin \left\{0, \frac{-2}{3}\right\}$, such that $8x^3 - 27x^6 + y^3 + 18x^3y = 0$, then the value of $\frac{2}{x} + \frac{y}{x^2}$ is equal to
2. In the figure, two concentric circles are given with radius of 10 cm & 5 cm. If A_1 & A_2 are area of sectors $\widehat{OPQ}$ & $\widehat{OAB}$ respectively, then $\frac{A_1}{A_2}$ is equal to



Space for Rough Work

3. The value of $\sqrt{31+10\sqrt{6}} - \sqrt{11+6\sqrt{2}} - \sqrt{4-2\sqrt{3}} - \sqrt{2.\sqrt{2}.\sqrt{2}.\sqrt{2}...\infty}$ is $(\sqrt{a}-1)(\sqrt{b}-1)$, then the value of ab is equal to
4. $A = \{2x : 1 \leq x \leq 50 \text{ \& } x \in \mathbb{N}\}$
 $B = \{3x : 1 \leq x \leq 33 \text{ \& } x \in \mathbb{N}\}$
 If C is a set of such elements of sets A & B which are divisible by 2 & 3 both, then the sum of digits of $n(C)$ is equal to
5. If $27^x - 2^2 \cdot 3^{2x} + 3^{x+1} = 0$, then the number of positive integral solution(s) is equal to
6. If $A = \left\{ x : \left(\frac{x-4}{x} \right)^{x-2} = \left(\frac{x}{x-4} \right)^{2x-3} \right\}$, then the sum of real values of all possible elements of the set A is λ , then the value of 3λ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator